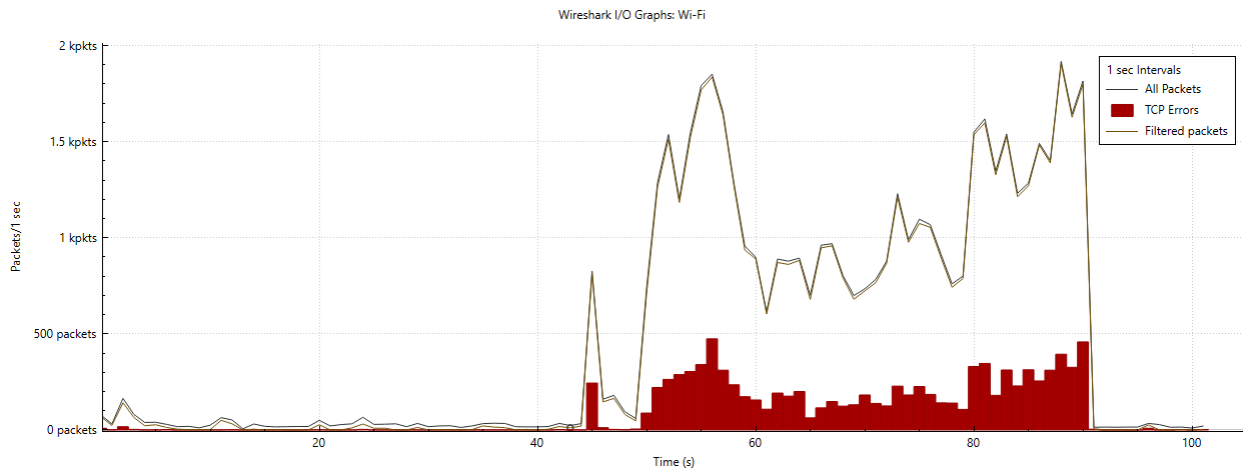


CNET Lab 9

Task 1



Bytes transmitted: 51800831

Duration: 1:41 = 101 seconds

1. Data Rate (Bit Rate) = Total bits transmitted / Transmission duration

Data Rate = $(51800831 \times 8) / (60 + 41) = 4.1 \text{ MBits/sec}$

2. Packet Transmission Rate (Packets/sec) = Total packets captured / Transmission duration

Packet Transmission Rate (Packets/sec) = $51221 / 101 = 507.13 \text{ packets/sec}$

3. Baud Rate (Signal Rate) using the formula given in the manual at the start.

Signal Rate (Baud) = Bit Rate (bps) / Bits per symbol (r)

Bits per symbol = 6

Baud rate = $4.1 \times 10^6 / 6 = 683.33 \times 10^3 \text{ baud}$

Q1. How does the observed data rate compare to the expected network speed? (hint: you will need to check your connection speed to see the expected network speed)

Expected network speed = $74.84 \text{ Mbps} = 74.84 \times 8 = 598.72 \text{ MBits/sec}$

Observed data rate = 4.1 MBits/sec

The large difference is due to using wireless network and the noise and interference caused by it.

Q2. If multiple bits per symbol are used, how does this affect the baud rate? (hint: you will have to find the modulation scheme for your network to find this)

If multiple bits per symbol are used, the higher bits per symbol will result in less baud rate. Because more bits per symbol will carry more information so less bits are needed for same bit rate.

Task 2:

Q1. How frequently do checksum errors occur in wired vs. wireless networks?

Wireless networks has much more frequency of checksum errors than wired network. Wired networks are stable so they barely have any checksum error like 1 in a million chance are there for them.

Q2. Why are Wi-Fi packets more prone to errors compared to Ethernet?

Wireless networks have more frequent checksum errors than wired network because wireless signals travel through the air and has more noise, and interference. Wired signals travel through shielded cables, so they are much more stable.